

# A variable-step Crank-Nicolson and high-order compact scheme for a nonlinear Black-Scholes type equation

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## Abstract

We study a variable-step Crank-Nicolson and high-order compact finite difference method for a nonlinear Black-Scholes type equation arising from option pricing with counterparty risk. By combining a variable-step Crank-Nicolson temporal approximation with a fourth-order compact spatial operator, and by introducing an explicit approximation for the nonlinear terms, we obtain a fully discrete linear scheme that only requires solving linear systems at each time level. The energy method is developed to show the unconditional stability and convergence of the proposed scheme, yielding the optimal error bound of  $\mathcal{O}(\tau^2 + h^4)$ , where  $\tau$  is the maximal time-step size and  $h$  is the uniform space-step size. Numerical experiments are presented to demonstrate the accuracy and efficiency.

**Keywords:** nonlinear Black-Scholes equation; variable-step method; Crank-Nicolson method; high-order convergence

**AMS subject classifications.** 65M06; 65M12; 35B65; 35R11

## 1 Introduction

In practical financial markets, derivative valuation is affected by funding costs [1], bilateral counterparty risk, recovery rates, and default intensities [8]. Duffie and Huang [6] pioneered the integration of credit risk into derivatives pricing, while Burgard and Kjaer [3, 4] extended the Black-Scholes framework to incorporate bilateral counterparty risk and funding costs, deriving nonlinear PDEs that capture the mutual default probabilities and nonlinear couplings between asset prices and credit spreads. These foundational works highlighted the need for robust numerical schemes to handle the inherent nonlinearities, such as jumping nonlinearities, in counterparty risk models.

Our work presents a highly accurate numerical method for solving the semilinear Black-Scholes equation with counterparty risk [3, 4]:

$$\frac{\partial U}{\partial \zeta} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 U}{\partial S^2} + \gamma S \frac{\partial U}{\partial S} - rU = -(1 - R_B)\lambda_B U^- + (1 - R_C)\lambda_C U^+ + s_F U^+ \quad (1.1)$$

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in  $(S, \zeta) \in (0, \infty) \times (0, T)$  with the final data  $U(S, T) = \Phi(S)$  given by the payoff function  $\Phi$ , where  $U(S, \zeta)$  is the adjusted price of a derivative contract under funding costs and bilateral counterparty risk,  $T$  is the expiry,  $r$  is the risk-free interest rate, and

$$u^+ := \max(u, 0) \quad \text{and} \quad u^- := \max(-u, 0)$$

which were often called jumping nonlinearities. The constants  $\gamma$  and  $\sigma$  are the effective financing rate and the volatility, respectively;  $\lambda_B$  and  $\lambda_C$  are the default intensities of the counterparties,  $s_F$  is the funding spread, and  $R_B, R_C \in [0, 1]$  are the recovery rates of the derivative positions.

Take  $a := \frac{1}{2}\sigma^2$  and  $\rho := \gamma - a$ . Applying the transformations  $t = T - \zeta$  and  $x = \ln S$  to (1.1), we obtain the following equivalent system

$$\frac{\partial}{\partial t}u + \mathcal{A}u = F(u) \quad \text{for} \quad (x, t) \in (-\infty, +\infty) \times (0, T), \quad (1.2)$$

subject to the initial condition  $u(x, 0) = \Phi(e^x) := \varphi(x)$ , where

$$\mathcal{A}u = -a \frac{\partial^2}{\partial x^2}u - \rho \frac{\partial}{\partial x}u + ru \quad \text{and} \quad F(u) = pu^- - qu^+,$$

provided that  $p := (1 - R_B)\lambda_B$  and  $q := (1 - R_C)\lambda_C + s_F$ .

There are very few studies on the numerical solution of the nonlinear Black-Scholes type equations with counterparty risk. In [2], a monotone iterative method was proposed for (1.2), which generates convergent solutions by transforming the semilinear PDEs into standard parabolic problems with monotone reaction functions. Another traditional approach in solving the relevant problems is the Monte Carlo method, e.g. [5], which is a stochastic numerical technique that estimates solutions by generating a large number of random samples to simulate the evolution of financial variables and average the results for reliable approximations. These methods often suffer from high computational costs and struggle with balancing efficiency and accuracy, limiting their applicability to real-time valuation. In this work, we propose a simple and highly accurate finite difference method for solving the governing nonlinear problem. We combine the Crank-Nicolson type approximation on nonuniform time grids with a spatial compact discretization, and derive an explicit time approximation for the jumping nonlinearities. The resulting implicit-explicit (IMEX) numerical scheme is linear on the nonuniform time grids while maintaining the high accuracy. It dynamically adapts the time step, addressing the situation where the analytical solution changes complexly with time, and will save considerable computational cost as no iterative method is required. The unconditional stability and high-order convergence are also rigorously derived by using energy method, where the optimal convergence accuracy of second-order in time and fourth-order in space are displayed. In addition, our numerical analysis permits a certain range of negative risk-free interest rate, since the near-zero or negative interest-rate regimes may also occur in practice [7], whereas the risk-free interest rate is often assumed to be positive in standard models.

The remainder of this paper is organized as follows. In Section 2, we combine a spatial fourth-order approximation with the variable-step Crank-Nicolson technique as well as a temporal second-order explicit approximation to construct a variable-step, high-order compact and linearized difference scheme. By using energy method, we derive the unconditional stability and high-order convergence of the proposed scheme. It permits a certain range of negative risk-free interest rate. Section 3 gives two numerical examples to illustrate the theoretical statements.

## 2 The variable-step high-order scheme

For the sake of simplicity and without loss of generality, we truncate the equation (1.2) on a bounded spatial domain  $\Omega := [0, L]$  and consider homogeneous boundary conditions, the non-homogeneous boundary case can be treated similarly via a simple function substitution.

Let  $M$  be a positive integer representing the number of spatial nodes. The spatial domain is discretized into a uniform grid  $\Omega_h := \{x_i = ih \mid i = 1, 2, \dots, M-1\}$  equipped with the step size  $h := L/M$ , and the closure  $\bar{\Omega}_h := \Omega_h \cup \{x_0 = 0, x_M = L\}$  includes the boundary points. The space of grid functions vanishing on the boundary is defined as  $\mathcal{V}_h := \{(v_i)_{i=0}^M \mid v_0 = v_M = 0\}$ . For any  $v, w \in \mathcal{V}_h$ , the discrete inner product and  $L^2$ -norm are given by  $\langle v, w \rangle_h := h \sum_{i=1}^M v_i w_i$  and  $\|v\|_h := \sqrt{\langle v, v \rangle_h}$ , respectively. The central difference operators are defined as

$$\delta_x^2 v_i := \frac{v_{i+1} - 2v_i + v_{i-1}}{h^2}, \quad \delta_{\hat{x}} v_i := \frac{v_{i+1} - v_{i-1}}{2h}, \quad i = 1, 2, \dots, M-1.$$

Let  $N$  be a positive integer representing the number of temporal nodes. We consider the possibly nonuniform time partition  $0 = t_0 < t_1 < \dots < t_N = T$ , and the time step sizes are given by  $\tau_n := t_n - t_{n-1}$  for  $n = 1, 2, \dots, N$ , and the maximum time step is denoted by  $\tau := \max_{1 \leq n \leq N} \{\tau_n\}$ . Define the adjacent step ratios by  $r_n := \tau_n / \tau_{n-1}$  ( $n = 2, 3, \dots, N$ ), and suppose that there exists a constant  $c_r \geq 1$  such that  $\max_{2 \leq n \leq N} \{r_n\} = c_r$ . Moreover, denote  $v^n$  the numerical approximation of  $v(t_n)$ , and define

$$v^{n-\frac{1}{2}} := \frac{v^n + v^{n-1}}{2} \quad \text{and} \quad \delta_\tau v^{n-\frac{1}{2}} := \frac{v^n - v^{n-1}}{\tau_n}.$$

### 2.1 The numerical scheme

To achieve a spatial high-order numerical scheme, we implement a fourth-order approximation technique discussed in [10]. Take

$$g(x, t) := \partial_t u(x, t) + ru(x, t) - F(u),$$

it holds that

$$\frac{h^2}{12} \left( \delta_x^2 g(x_i, t) + \frac{\rho}{a} \delta_{\hat{x}} g(x_i, t) \right) + g(x_i, t) = \left( a + \frac{h^2 \rho^2}{12a} \right) \delta_x^2 u(x_i, t) + \rho \delta_{\hat{x}} u(x_i, t) + \mathcal{O}(h^4), \quad (2.1)$$

for  $1 \leq i \leq M-1$ . The detailed derivation of the above formula can be found in [9, Subsection 2.1].

The following lemma, which can be checked easily by using the Taylor formula, will be used to generate an explicit approximation of some terms in the governing model.

**Lemma 2.1.** *Let  $v(t) \in \mathcal{C}^3(t_{n-2}, t_n)$ ,  $n \geq 2$ . It holds that*

$$v(t_{n-\frac{1}{2}}) = \left( \frac{1}{2} r_n + 1 \right) v(t_{n-1}) - \frac{1}{2} r_n v(t_{n-2}) + \mathcal{O}(\tau_n^2), \quad n \geq 2.$$

Define the difference operator  $\mathcal{H} := \frac{h^2}{12} (\delta_x^2 + \frac{\rho}{a} \delta_{\hat{x}}) + 1$ , and denote

$$\bar{u}(x_i, t_{n-\frac{1}{2}}) := \left( \frac{1}{2} r_n + 1 \right) u(x_i, t_{n-1}) - \frac{1}{2} r_n u(x_i, t_{n-2}) \quad \text{if } n \geq 2,$$

and  $\bar{u}(x_i, t_{\frac{1}{2}}) := u(x_i, t_0)$ .

Consider the equations in (2.1) at the time points  $t_{n-1/2}$  ( $1 \leq n \leq N$ ). Applying the explicit approximation established in Lemma 2.1 to the terms in  $F(u)$ , we obtain the Crank-Nicolson type IMEX and linear approximation

$$\begin{aligned} & \mathcal{H} \left[ \delta_\tau u(x_i, t_{n-\frac{1}{2}}) + r \frac{u(x_i, t_n) + u(x_i, t_{n-1})}{2} - F(\bar{u}(x_i, t_{n-\frac{1}{2}})) \right] \\ &= \left( a + \frac{h^2 \rho^2}{12a} \right) \delta_x^2 \frac{u(x_i, t_n) + u(x_i, t_{n-1})}{2} + \rho \delta_{\hat{x}} \frac{u(x_i, t_n) + u(x_i, t_{n-1})}{2} + R_i^{n-\frac{1}{2}}, \end{aligned} \quad (2.2)$$

where  $R_i^{n-\frac{1}{2}} := \mathcal{H}(R_{t_1})_i^{n-\frac{1}{2}} + \mathcal{H}(R_{t_2})_i^{n-\frac{1}{2}} + (R_{t_3})_i^{n-\frac{1}{2}} + (R_S)_i^{n-\frac{1}{2}}$ , in which  $(R_S)_i^{n-\frac{1}{2}} = \mathcal{O}(h^4)$  and

$$\begin{aligned} (R_{t_1})_i^{n-\frac{1}{2}} &:= \delta_\tau u_i^{n-\frac{1}{2}} - \partial_t u(x_i, t_{n-\frac{1}{2}}) + r \hat{R}_i^{n-\frac{1}{2}} \text{ with } \hat{R}_i^{n-\frac{1}{2}} := \frac{u(x_i, t_n) + u(x_i, t_{n-1})}{2} - u(x_i, t_{n-\frac{1}{2}}), \\ (R_{t_2})_i^{n-\frac{1}{2}} &:= F(\bar{u}(x_i, t_{n-\frac{1}{2}})) - F(u(x_i, t_{n-\frac{1}{2}})), \\ (R_{t_3})_i^{n-\frac{1}{2}} &:= \left[ \left( a + \frac{h^2 \rho^2}{12a} \right) \delta_x^2 + \rho \delta_{\hat{x}} \right] \hat{R}_i^{n-\frac{1}{2}}. \end{aligned}$$

It is easy to see that  $R_i^{n-\frac{1}{2}} = \mathcal{O}(\tau_n^2 + h^4)$  for  $n \geq 2$ , and  $R_i^{\frac{1}{2}} = \mathcal{O}(\tau_1 + h^4)$ .

Let  $u_i^n$  be the numerical approximations of  $u(x_i, \tau_n)$  for  $x_i \in \Omega_h$ ,  $0 \leq n \leq N$ , and denote

$$\bar{u}_i^{n-\frac{1}{2}} := \left( \frac{1}{2} r_n + 1 \right) u_i^{n-1} - \frac{1}{2} r_n u_i^{n-2} \quad \text{and} \quad F_i^{n-\frac{1}{2}} = F(\bar{u}_i^{n-\frac{1}{2}}), \quad n \geq 1.$$

Omitting the truncation error in (2.2), we obtain the following numerical scheme

$$\mathcal{H} \left( \delta_\tau u_i^{n-\frac{1}{2}} + r u_i^{n-\frac{1}{2}} - F_i^{n-\frac{1}{2}} \right) = \left( a + \frac{h^2 \rho^2}{12a} \right) \delta_x^2 u_i^{n-\frac{1}{2}} + \rho \delta_{\hat{x}} u_i^{n-\frac{1}{2}}, \quad (2.3)$$

for  $n \geq 1$  and  $1 \leq i \leq M-1$ ; equipped with the initial condition:  $u_i^0 = \varphi_i$  for  $x_i \in \Omega_h$  and the zero boundary conditions:  $u_0^n = u_M^n = 0$ .

Denote the vectors

$$\mathbf{u}^n := [u_1^n, u_2^n, \dots, u_{M-1}^n]^T, \quad \mathbf{F}^{n-\frac{1}{2}} := [F_1^{n-\frac{1}{2}}, F_2^{n-\frac{1}{2}}, \dots, F_{M-1}^{n-\frac{1}{2}}]^T,$$

and the  $(M-1)$ -dimensional tridiagonal matrices

$$A := \text{tridiag}(1, -2, 1), \quad S := \text{tridiag}(-1, 0, 1).$$

Take

$$H := \frac{1}{12} A + \frac{h\rho}{24a} S + I$$

with  $I$  being the  $(M-1)$ -dimensional identity matrix. Then the scheme (2.3) can be written in the matrix-vector form

$$H \left( \delta_\tau \mathbf{u}^{n-\frac{1}{2}} + r \mathbf{u}^{n-\frac{1}{2}} - \mathbf{F}^{n-\frac{1}{2}} \right) = \left( \frac{a}{h^2} + \frac{\rho^2}{12a} \right) A \mathbf{u}^{n-\frac{1}{2}} + \frac{\rho}{2h} S \mathbf{u}^{n-\frac{1}{2}}, \quad n \geq 1, \quad (2.4)$$

subject to the initial condition  $\mathbf{u}^0 := [\varphi_1, \varphi_2, \dots, \varphi_{M-1}]^T$  and the zero boundary conditions.

## 2.2 Stability and convergence

**Lemma 2.2.** [9, Lemma 3.4] For all  $\mathbf{w} \in \mathbb{R}^{M-1}$ , the matrix  $H^T H$  satisfies

$$\frac{5}{12} \mathbf{w}^T \mathbf{w} \leq \mathbf{w}^T H^T H \mathbf{w} \leq \mathbf{w}^T \mathbf{w}.$$

**Lemma 2.3.** [9, Lemma 3.5] The matrix  $H^T S + SH$  is negative semi-definite.

To process the numerical analysis of the proposed scheme, we also need to bound the numerical range of the matrix  $H^T A + AH$  which can be divided as

$$H^T A + AH = \frac{1}{6} P + D,$$

where  $P$  is an  $(M-1) \times (M-1)$  symmetric pentadiagonal matrix

$$P := \begin{bmatrix} -18 & 8 & 1 & & & & \\ 8 & -18 & 8 & 1 & & & \\ 1 & 8 & -18 & 8 & 1 & & \\ & \ddots & \ddots & \ddots & \ddots & \ddots & \\ & & 1 & 8 & -18 & 8 & 1 \\ & & & 1 & 8 & -18 & 8 \\ & & & & 1 & 8 & -18 \end{bmatrix}$$

and  $D$  is a diagonal matrix

$$D := \begin{bmatrix} 1 - \frac{\rho}{2a}h & & & & & \\ & 0 & & & & \\ & & \ddots & & & \\ & & & 0 & & \\ & & & & -1 + \frac{\rho}{2a}h & \end{bmatrix}.$$

Take  $\theta_k := \frac{k\pi}{M}$  for  $k = 1, 2, \dots, M-1$ . The eigenvalues of  $P$  are given in the following lemma.

**Lemma 2.4.** The eigenvalues of  $P$  are given by

$$\lambda_k = -18 + 16 \cos(\theta_k) + 2 \cos(2\theta_k), \quad k = 1, \dots, M-1.$$

*Proof.* Denote the vector

$$\mathbf{v}^{(k)} := [\sin(\theta_k) \quad \sin(2\theta_k) \quad \cdots \quad \sin((M-1)\theta_k)]^T, \quad k = 1, \dots, M-1.$$

Consider the  $i$ -th ( $1 \leq i \leq M-1$ ) entry of the vector  $P\mathbf{v}^{(k)}$ , we have

$$\begin{aligned} (P\mathbf{v}^{(k)})_i &= \sum_{j=1}^{M-1} P_{ij} \sin(j\theta_k) \\ &= [\sin((i-2)\theta_k) + \sin((i+2)\theta_k)] + 8 [\sin((i-1)\theta_k) + \sin((i+1)\theta_k)] - 18 \sin(i\theta_k) \\ &= 2 \sin(i\theta_k) \cos(2\theta_k) + 16 \sin(i\theta_k) \cos(\theta_k) - 18 \sin(i\theta_k) \\ &= [-18 + 16 \cos(\theta_k) + 2 \cos(2\theta_k)] \sin(i\theta_k). \end{aligned}$$

It follows that

$$P\mathbf{v}^{(k)} = \lambda_k \mathbf{v}^{(k)}, \quad k = 1, \dots, M-1.$$

□

**Lemma 2.5.** *For any  $\mathbf{w} \in \mathbb{R}^{M-1}$ . It holds that*

$$\mathbf{w}^T (H^T A + AH) \mathbf{w} \leq -\frac{5\pi^2}{3L^2} h^2 \mathbf{w}^T \mathbf{w}.$$

*Proof.* Notice that

$$\theta_k \geq \theta_1 = \frac{\pi}{M} = \frac{\pi}{L} h \quad \text{for } k = 1, 2, \dots, M-1.$$

Then, according to Lemma 2.4, the maximum eigenvalue of  $P$  is  $\lambda_1$ . On the other hand, one can use Taylor expansions to obtain

$$\cos(\theta_1) \leq 1 - \frac{\theta_1^2}{2} + \frac{\theta_1^4}{24}, \quad \cos(2\theta_1) \leq 1 - 2\theta_1^2 + \frac{2\theta_1^4}{3}.$$

Thus, the maximum eigenvalue of  $P$  satisfies

$$\begin{aligned} \lambda_{\max}(P) = \lambda_1 &\leq -18 + 16 \left( 1 - \frac{\theta_1^2}{2} + \frac{\theta_1^4}{24} \right) + 2 \left( 1 - 2\theta_1^2 + \frac{2\theta_1^4}{3} \right) \\ &= -12\theta_1^2 + 2\theta_1^4 \\ &\leq -12\theta_1^2 + 2\theta_1^2 \\ &= -\frac{10\pi^2}{L^2} h^2, \end{aligned}$$

where the fact  $\theta_1 < 1$  (for  $M > 3$ ) has been utilized.

Moreover, it is obviously that  $D$  is negative semi-definite for small  $h$ , then

$$\lambda_{\max}(H^T A + AH) \leq \frac{1}{6} \lambda_{\max}(P) \leq -\frac{5\pi^2}{3L^2} h^2.$$

□

We also need the following discrete grönwall inequality in the proofs of stability and convergence.

**Lemma 2.6.** *[11] Let nonnegative sequences  $\{v_n\}$  and  $\{w_n\}$  satisfy*

$$v_n \leq G + \sum_{k=0}^{n-1} w_k v_k, \quad n \geq 1,$$

*with  $G \geq 0$ . Then*

$$v_n \leq G \exp \left( \sum_{k=0}^{n-1} w_k \right).$$

Define the norm  $\|u\|_H^2 := h\mathbf{u}^T H^T H \mathbf{u}$ , it follows from Lemma 2.2 that

$$\frac{5}{12} \|u\|^2 \leq \|u\|_H^2 \leq \|u\|^2. \quad (2.5)$$

We have the following stability result.

**Theorem 2.7.** Suppose  $r \geq -\frac{5a\pi^2}{6L^2}$ . The proposed scheme (2.3) is unconditionally stable, and

$$\|u^n\|_H \leq \exp((c_1 + c_2)t_n) \|u^0\|_H, \quad 1 \leq n \leq N. \quad (2.6)$$

where  $c_1 := \frac{1}{2}(|p| + |q|)(\frac{1}{2}c_r + 1)$  and  $c_2 := \frac{1}{4}(|p| + |q|)c_r$ . Moreover,

$$\|u^n\| \leq \frac{12}{5} \exp((c_1 + c_2)t_n) \|u^0\|, \quad 1 \leq n \leq N. \quad (2.7)$$

*Proof.* Multiplying both sides of (2.4) with  $h(H\mathbf{u}^{n-\frac{1}{2}})^T$ , one has

$$\begin{aligned} & h(\mathbf{u}^{n-\frac{1}{2}})^T H^T H \delta_\tau \mathbf{u}^{n-\frac{1}{2}} + rh(\mathbf{u}^{n-\frac{1}{2}})^T H^T H \mathbf{u}^{n-\frac{1}{2}} - h(\mathbf{u}^{n-\frac{1}{2}})^T H^T H \mathbf{F}^{n-\frac{1}{2}} \\ &= \left( \frac{a}{h^2} + \frac{\rho^2}{12a} \right) h(\mathbf{u}^{n-\frac{1}{2}})^T H^T A \mathbf{u}^{n-\frac{1}{2}} + \frac{\rho}{2h} h(\mathbf{u}^{n-\frac{1}{2}})^T H^T S \mathbf{u}^{n-\frac{1}{2}}. \end{aligned} \quad (2.8)$$

According to Lemmas 2.3 and 2.5, the terms on the right-hand side of (2.8) can be bounded as

$$\begin{aligned} & \left( \frac{a}{h^2} + \frac{\rho^2}{12a} \right) h(\mathbf{u}^{n-\frac{1}{2}})^T H^T A \mathbf{u}^{n-\frac{1}{2}} + \frac{\rho}{2h} h(\mathbf{u}^{n-\frac{1}{2}})^T H^T S \mathbf{u}^{n-\frac{1}{2}} \\ &= \frac{h}{2} \left( \frac{a}{h^2} + \frac{\rho^2}{12a} \right) (\mathbf{u}^{n-\frac{1}{2}})^T (H^T A + A H) \mathbf{u}^{n-\frac{1}{2}} + \frac{\rho}{4} (\mathbf{u}^{n-\frac{1}{2}})^T (H^T S + S^T H) \mathbf{u}^{n-\frac{1}{2}} \\ &\leq -\frac{5\pi^2}{6L^2} \left( a + \frac{\rho^2 h^2}{12a} \right) \|u^{n-\frac{1}{2}}\|^2 \\ &< -\frac{5a\pi^2}{6L^2} \|u^{n-\frac{1}{2}}\|^2. \end{aligned} \quad (2.9)$$

For the terms on the left hand side of (2.8), direct calculations give

$$h(\mathbf{u}^{n-\frac{1}{2}})^T H^T H \delta_\tau \mathbf{u}^{n-\frac{1}{2}} = \frac{\|u^n\|_H^2 - \|u^{n-1}\|_H^2}{2\tau_n}, \quad rh(\mathbf{u}^{n-\frac{1}{2}})^T H^T H \mathbf{u}^{n-\frac{1}{2}} = r\|u^{n-\frac{1}{2}}\|_H^2. \quad (2.10)$$

Using the Cauchy-Schwarz inequality and the triangular inequality, we have

$$\begin{aligned} h(\mathbf{u}^{n-\frac{1}{2}})^T H^T H \mathbf{F}^{n-\frac{1}{2}} &\leq \|u^{n-\frac{1}{2}}\|_H \|F^{n-\frac{1}{2}}\|_H \\ &\leq \begin{cases} c_1 (\|u^1\|_H + \|u^0\|_H) \|u^0\|_H, & \text{for } n = 1, \\ (\|u^n\|_H + \|u^{n-1}\|_H) (c_1 \|u^{n-1}\|_H + c_2 \|u^{n-2}\|_H), & \text{for } n \geq 2, \end{cases} \end{aligned} \quad (2.11)$$

where the following inequality has been applied

$$\|F^{n-\frac{1}{2}}\|_H \leq (|p| + |q|) \|\bar{u}^{n-\frac{1}{2}}\|_H \leq \begin{cases} (|p| + |q|) \|u^0\|_H, & \text{for } n = 1, \\ (|p| + |q|) \left[ (\frac{1}{2}c_r + 1) \|u^{n-1}\|_H + \frac{1}{2}c_r \|u^{n-2}\|_H \right], & \text{for } n \geq 2. \end{cases}$$

Combining (2.8)–(2.11) and using Lemma 2.2, and with  $r \geq -\frac{5a\pi^2}{6L^2}$ , we get

$$\|u^n\|_H^2 - \|u^{n-1}\|_H^2 \leq \begin{cases} c_1 \tau_1 (\|u^1\|_H + \|u^0\|_H) \|u^0\|_H, & \text{for } n = 1, \\ \tau_n (\|u^n\|_H + \|u^{n-1}\|_H) (c_1 \|u^{n-1}\|_H + c_2 \|u^{n-2}\|_H), & \text{for } n \geq 2. \end{cases} \quad (2.12)$$

If  $\|u^n\|_H \leq \|u^{n-1}\|_H$ , the desired result (2.6) holds obviously. If  $\|u^n\|_H > \|u^{n-1}\|_H$ , it follows from (2.12) that

$$\|u^1\|_H - \|u^0\|_H \leq c_1 \tau_1 \|u^0\|_H,$$

which confirms (2.6) for  $n = 1$ , and

$$\|u^n\|_H - \|u^{n-1}\|_H \leq \tau_n (c_1 \|u^{n-1}\|_H + c_2 \|u^{n-2}\|_H) \quad \text{for } n \geq 2.$$

We obtain that

$$\|u^n\|_H \leq \|u^0\|_H + (c_1 + c_2) \sum_{k=0}^{n-1} \tau_{k+1} \|u^k\|_H, \quad n \geq 2.$$

Applying the Grönwall inequality in Lemma 2.6, we get

$$\|u^n\|_H \leq \exp((c_1 + c_2)t_n) \|u^0\|_H, \quad n \geq 2.$$

Thus, (2.6) is valid for  $n \geq 1$ . Combined with (2.5), the result (2.7) can be obtained.  $\square$

Denote  $e_i^k := u(x_i, t_k) - u_i^k$ . We then show the unconditional convergence of the proposed scheme.

**Theorem 2.8.** *Suppose  $r \geq -\frac{5a\pi^2}{6L^2}$ . The scheme (2.3) is unconditionally convergent and*

$$\|e^n\|_H \leq C(\tau^2 + h^4), \quad 1 \leq n \leq N.$$

Moreover,

$$\|e^n\| \leq C(\tau^2 + h^4), \quad 1 \leq n \leq N.$$

*Proof.* We have the following error equation

$$H \left( \delta_\tau \mathbf{e}^{n-\frac{1}{2}} + r \mathbf{e}^{n-\frac{1}{2}} - \hat{\mathbf{F}}^{n-\frac{1}{2}} \right) = \left( \frac{a}{h^2} + \frac{\rho^2}{12a} \right) A \mathbf{e}^{n-\frac{1}{2}} + \frac{\rho}{2h} S \mathbf{e}^{n-\frac{1}{2}} + \mathbf{R}^{n-\frac{1}{2}}, \quad (2.13)$$

where

$$\begin{aligned} \mathbf{e}^{n-\frac{1}{2}} &:= [e_1^{n-\frac{1}{2}}, e_2^{n-\frac{1}{2}}, \dots, e_{M-1}^{n-\frac{1}{2}}]^T, \\ \mathbf{R}^{n-\frac{1}{2}} &:= [R_1^{n-\frac{1}{2}}, R_2^{n-\frac{1}{2}}, \dots, R_{M-1}^{n-\frac{1}{2}}]^T, \\ \hat{\mathbf{F}}^{n-\frac{1}{2}} &:= [\hat{F}_1^{n-\frac{1}{2}}, \hat{F}_2^{n-\frac{1}{2}}, \dots, \hat{F}_{M-1}^{n-\frac{1}{2}}]^T \text{ with } \hat{F}_i^{k-\frac{1}{2}} := F(\bar{u}(x_i, t_{k-\frac{1}{2}})) - F_i^{k-\frac{1}{2}}. \end{aligned}$$

Multiplying both sides of (2.13) with  $h(H\mathbf{e}^{n-\frac{1}{2}})^T$ , and by using analogous derivations to the proof of Theorem 2.7, one can easily obtain that

$$\|e^1\|_H - \|e^0\|_H \leq c_1 \tau_1 (\|e^0\|_H + \|R^{\frac{1}{2}}\|_H)$$

which leads to

$$\|e^1\|_H \leq c_1 \tau_1 \|R^{\frac{1}{2}}\|_H \leq C(\tau_1^2 + h^2),$$

and

$$\|e^n\|_H \leq \|e^1\|_H + (c_1 + c_2) \sum_{k=1}^{n-1} \tau_{k+1} \|e^k\|_H + \sum_{k=2}^n \tau_{k+1} \|R^{k-\frac{1}{2}}\|_H \quad \text{for } n \geq 2.$$

Then, by the Grönwall inequality in Lemma 2.6, we get

$$\|e^n\|_H \leq C(\|e^1\|_H + \max_{2 \leq k \leq n} \|R^{k-\frac{1}{2}}\|_H) \leq C(\tau^2 + h^2), \quad n \geq 2.$$

The result (2.7) can be obtained by (2.5).  $\square$

### 3 Numerical implementation

In this section, we present two numerical examples to illustrate the performance of the proposed method. The first example is designed to verify the temporal second-order and spatial fourth-order accuracies. In this case, the spatial interval is divided uniformly into  $M$  subintervals, while the time interval  $[0, T]$  is discretized by a graded mesh

$$t_k = T \left( \frac{k}{N} \right)^{r_{\text{grade}}}, \quad k = 0, 1, \dots, N.$$

The second example is a no-exact-solution test based on European option payoffs. In this case, the computation is performed in the log-price variable  $x = \ln S$ , and the numerical results are displayed with respect to the underlying asset price  $S$ . To better resolve the possible non-smoothness of the initial payoff near  $t = 0$ , we use a piecewise temporal mesh: a graded mesh is adopted on the initial subinterval  $[0, T_0]$ , followed by a uniform mesh on  $[T_0, T]$ .

**Example 3.1.** *We consider problem (1.2) (by adding a source term  $f(x, t)$  on its right-hand side) on  $\Omega = (0, 1)$  with terminal time  $T = 1$ . The initial condition is  $u(x, 0) = -\frac{1}{2}x^3(1-x)^3$  and the source term is chosen as*

$$f(x, t) = \frac{3}{2}t^{1/2}(x^3 - 3x^4 + 3x^5 - x^6) + (t^{3/2} - \frac{1}{2})(-0.12x + 0.705x^2 - 1.145x^3 + 0.540x^4 + 0.015x^5 + 0.005x^6) - 0.01 \left[ x^3(1-x)^3(t^{3/2} - \frac{1}{2}) \right]^- + 0.022 \left[ x^3(1-x)^3(t^{3/2} - \frac{1}{2}) \right]^+.$$

where the parameters are set as

$$\sigma = 0.2, r = -0.005, \gamma = 0.025, R_B = 0.5, R_C = 0.6, \lambda_B = 0.02, \lambda_C = 0.03 \text{ and } s_F = 0.01.$$

The exact solution of this example is  $u(x, t) = x^3(1-x)^3(t^{3/2} - \frac{1}{2})$ .

Here we take  $r_{\text{grade}} = 2$ . Denote  $E_2(M, N) := \max_{1 \leq n \leq N} \|u(\cdot, t_n) - u^n\|$ , the temporal and spatial convergence rates are computed, respectively, by

$$\text{Rate}_\tau = \log_2 \left[ \frac{E_2(M, N)}{E_2(M, 2N)} \right], \quad \text{Rate}_h = \log_2 \left[ \frac{E_2(M, N)}{E_2(2M, N)} \right].$$

From Tables 1 and 2, the temporal second-order and the spatial fourth-order accuracies of the proposed scheme are well demonstrated.

Table 1: Temporal accuracy with fixed  $M = 100$ .

| $N$               | $E_2(M, N)$             | $\text{Rate}_\tau$ |
|-------------------|-------------------------|--------------------|
| 10                | $7.6580 \times 10^{-6}$ | *                  |
| 20                | $1.9709 \times 10^{-6}$ | 2.03               |
| 40                | $4.9999 \times 10^{-7}$ | 2.02               |
| 80                | $1.2608 \times 10^{-7}$ | 2.01               |
| Theoretical Order |                         | 2.00               |

Table 2: Spatial accuracy with fixed  $N = 800$ .

| $M$               | $E_2(M, N)$             | Rate <sub><math>h</math></sub> |
|-------------------|-------------------------|--------------------------------|
| 4                 | $3.7496 \times 10^{-5}$ | *                              |
| 8                 | $2.3742 \times 10^{-6}$ | 3.98                           |
| 16                | $1.4760 \times 10^{-7}$ | 4.01                           |
| 32                | $8.2881 \times 10^{-9}$ | 4.15                           |
| Theoretical Order |                         | 4.00                           |

**Example 3.2.** We next consider a no-exact-solution example based on European option payoffs. The computation is carried out in the log-price variable  $x = \ln S$ , where the underlying asset price  $S \in [0.001, 20]$ . The terminal time is set as  $T = 1$ , and the model parameters are chosen as

$$\sigma = 0.25, r = 0.05, \gamma = 0.03, R_B = 0.40, R_C = 0.50, \lambda_B = 0.02, \lambda_C = 0.03 \text{ and } s_F = 0.02.$$

We take the strike price as  $K = 10$  and the fixed cash deduction as  $C = \eta K$ , where  $\eta = 0.15$ . Thus,  $C = 1.5$ .

For the European call case, the initial condition is given by the cash-adjusted call payoff

$$u(x, 0) = \max(e^x - K, 0) - C.$$

For the European put case, the initial condition is given by the cash-adjusted put payoff

$$u(x, 0) = \max(K - e^x, 0) - C.$$

The settings in Example 3.2 preserve the standard European call and put structures, while the fixed cash deduction naturally yields both positive and negative value regions. Therefore, both nonlinear components  $u^+$  and  $u^-$  are involved in the computation.

In the numerical implementation, we take  $M = 100$  and  $N = 100$ . The time interval  $[0, T]$  is divided into two parts. On the initial interval  $[0, T_0]$ , we use a graded mesh

$$t_k = T_0 \left( \frac{k}{N_0} \right)^{r_{\text{grade}}}, k = 0, 1, \dots, N_g,$$

where  $T_0 = 0.2T$ ,  $N_0 = 20$ , and  $r_{\text{grade}} = 2$ . On the remaining interval  $[T_0, T]$ , we use a uniform mesh

$$t_{N_0+j} = T_0 + \frac{j}{N_u}(T - T_0), j = 1, 2, \dots, N_u,$$

where  $N_u = 80$ .

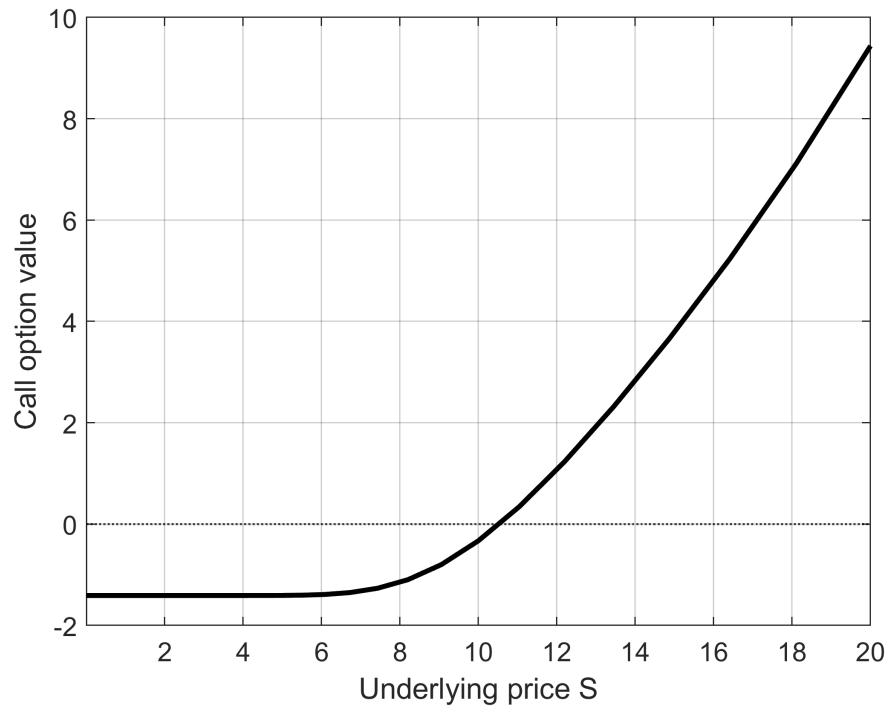


Figure 1: Numerical solution for the European call option at  $t = T$ .

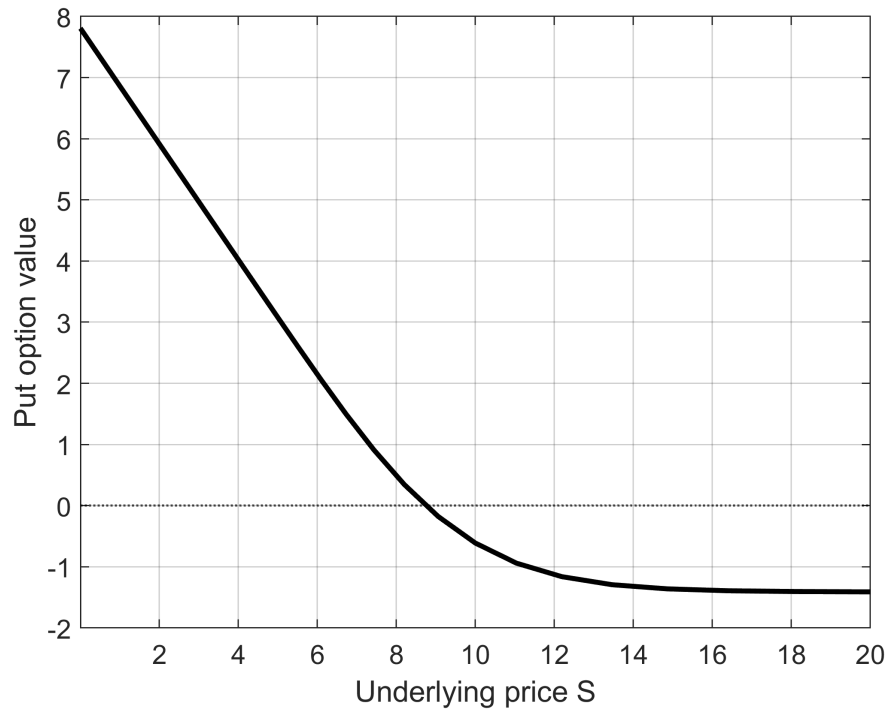


Figure 2: Numerical solution for the European put option at  $t = T$ .

Figures 1 and 2 show the numerical solutions of the proposed scheme for the European call and put options, respectively, illustrating the applicability of the proposed method to the practical option-pricing problems.

## References

- [1] R. F. Almgren, Optimal execution with nonlinear impact functions and trading-enhanced risk, *Appl. Math. Finance*, 10 (2005), 1–18.
- [2] B. Alziary, P. Takáč, Monotone methods in counterparty risk models with nonlinear Black–Scholes-type equations, *SeMA J.*, 80 (2023), 353–379.
- [3] C. Burgard, M. Kjaer, Addendum to: “PDE representations of derivatives with bilateral counterparty risk and funding costs”, *SSRN Electron. J.*, 2109723 (2012).
- [4] C. Burgard, M. Kjaer, Partial differential equation representations of derivatives with bilateral counterparty risk and funding costs, *J. Credit Risk*, 7 (2011), 1–19.
- [5] S. Crépey, T. M. Nguyen, Nonlinear Monte Carlo schemes for counterparty risk on credit derivatives, *Springer Proc. Math. Stat.*, 165 (2016), 53–82.
- [6] D. Duffie, M. Huang, Swap rates and credit quality, *J. Finance*, 51 (1996), 921–949.
- [7] C. Grisse, S. Schumacher, The response of long-term yields to negative interest rates: evidence from Switzerland, *SNB Working Papers*, No. 2017-10 (2017).
- [8] R. C. Merton, Option pricing when underlying stock returns are discontinuous, *J. Financ. Econ.*, 3 (1976), 125–144.
- [9] K. Song, P. Lyu, A high-order and fast scheme with variable time steps for the time-fractional Black-Scholes equation, *Math. Methods Appl. Sci.*, 46 (2023), 1990–2011.
- [10] R. de Staelen, A. Hendy, Numerically pricing double barrier options in a time-fractional Black-Scholes model, *Comput. Math. Appl.*, 74 (2017), 1166–1175.
- [11] J. W. Thomas, *Numerical Partial Differential Equations: Finite Difference Methods*, Springer, New York, 1995.